

EduGenie-AI

Requirement Analysis Phase

Google Gemini Powered AI Learning Assistant

Page 1 – Introduction and Requirement Overview

Introduction

Requirement Analysis is the second major phase of the EduGenie-AI project. It converts the idea developed during brainstorming into clear and measurable requirements. The purpose is to identify what the system must provide to students, how the application should behave, and what technical resources are necessary.

Project Requirement Goal

EduGenie-AI is planned as a Google Gemini-powered AI learning assistant for college students. The requirement analysis focuses on creating a simple web application that can answer academic questions, explain concepts, generate quizzes, summarize learning content, and recommend learning topics.

Stakeholder

The main stakeholder is the student or learner who interacts with the application. Other stakeholders can include teachers, project evaluators, developers, and administrators who may be interested in the educational usefulness, reliability, maintainability, and security of the system.

User Expectations

The user expects the system to be easy to operate. A student should be able to enter a question or topic, select an appropriate learning function, and receive a meaningful response without needing technical knowledge.

Requirement Categories

The requirements are organized into functional requirements, non-functional requirements, software requirements, hardware requirements, interface requirements, security requirements, and testing requirements. This classification makes the project easier to design and verify.

EduGenie-AI

Requirement Analysis Phase

Google Gemini Powered AI Learning Assistant

Page 2 – Functional Requirements

Question and Answer Requirement

The system shall provide a question-answering function. The student can enter an academic question and the application shall send the request to the backend, which prepares an appropriate prompt for Google Gemini and returns the generated answer.

Explanation Requirement

The system shall provide an explanation feature. A student can enter a difficult concept or topic and receive a detailed explanation written in simple language. The response may include definitions, important points, examples, and step-by-step descriptions.

Quiz Requirement

The system shall provide quiz generation. A student can provide a topic and request practice questions. The generated quiz can be used for self-assessment and revision.

Summarization Requirement

The system shall accept learning content and generate a shorter summary containing the major points. This function is intended to support examination preparation and quick revision.

Learning Recommendation Requirement

The system shall provide learning recommendations based on a topic or learning goal. The response may suggest related concepts or a sequence of topics that a student can study next.

Health and API Requirements

The FastAPI application shall provide a health/status route to verify that the service is running. Separate routes shall be available for the main learning functions.

EduGenie-AI

Requirement Analysis Phase

Google Gemini Powered AI Learning Assistant

Page 3 – Non-Functional and Technical Requirements

Usability Requirements

The interface should be simple, clear, and student-friendly. Buttons, input fields, labels, and results should be understandable without technical knowledge. The design should avoid unnecessary complexity.

Performance Requirements

The application should process requests efficiently within the limits of the external Gemini API. The system should provide useful error messages if an AI request fails or the service is temporarily unavailable.

Reliability Requirements

The application should handle common errors without crashing. Missing inputs, incorrect configuration, API failures, and unexpected responses should be handled in a controlled way.

Maintainability Requirements

The project should use a modular structure. Separate Python files can manage Q&A, explanations, quizzes, summaries, and learning recommendations. This makes individual features easier to modify and test.

Software Requirements

The planned software stack includes Python, FastAPI, Uvicorn, Google GenAI SDK, python-dotenv, Jinja2 templates, HTML, CSS, JavaScript, Visual Studio Code, Git, and GitHub.

Hardware and Network Requirements

A standard laptop or desktop computer is sufficient for development. An internet connection is required because the application communicates with the Gemini service through an API.

EduGenie-AI

Requirement Analysis Phase

Google Gemini Powered AI Learning Assistant

Page 4 – Security, Interface and Data Requirements

API Key Security

The Gemini API key is a sensitive credential and must not be placed directly in frontend files or public source code. The application should load the key from an environment file or environment variable.

Git Security

The .env file should be excluded using .gitignore so that the API key is not accidentally committed to a Git repository. Developers should verify Git status before committing project files.

Frontend Requirement

The frontend shall provide an easy way for users to enter questions, topics, or content and select the required learning function. Results should be displayed clearly on the page.

Backend Requirement

FastAPI shall receive frontend requests, connect each request to the correct learning module, and return the generated result to the browser.

Data Flow Requirement

The basic data flow is: student input → JavaScript request → FastAPI route → learning module → Gemini API → generated response → FastAPI → browser.

Response Quality

Prompts should instruct Gemini to provide clear, educational, and appropriately detailed answers. Because generated AI responses can contain errors, students should verify important academic information using trusted learning resources.

EduGenie-AI

Requirement Analysis Phase

Google Gemini Powered AI Learning Assistant

Page 5 – Requirement Specification and Conclusion

Minimum Viable Requirements

The minimum working version of EduGenie-AI should include the web interface, FastAPI backend, Gemini integration, Q&A, explanation, quiz, summarization, and learning recommendation features.

Testing Requirements

Every major function should be tested using normal inputs and different academic topics. Empty or invalid inputs should also be tested. Frontend-to-backend integration must be verified to ensure that requests and responses move correctly through the system.

Documentation Requirements

The project documentation should include installation instructions, software requirements, configuration steps, system architecture, module descriptions, API routes, feature descriptions, testing results, limitations, and future enhancements.

Acceptance Criteria

The system can be considered ready for the next development stage when the application starts successfully, the main interface loads, the core API routes respond, Gemini integration works, API credentials remain protected, and the primary learning functions can be demonstrated.

Conclusion

Requirement Analysis provides a clear technical and functional foundation for EduGenie-AI. By defining user needs, core functions, quality expectations, security requirements, technologies, and testing criteria, the project can proceed to the Project Design phase with a well-defined scope.